

GROUP-FAIR STOCHASTIC GRADIENT DESCENT VIA MASKED OPTIMAL TRANSPORT

Herlock (SeyedAbolfazl) Rahimi Dionysis Kalogerias

Department of Electrical and Computer Engineering, Yale University, New Haven, CT, USA
{herlock.rahimi, dionysis.kalogerias}@yale.edu

ABSTRACT

Minibatch SGD is group-blind. Which groups appear in a batch is decided by how often they occur in the data, not by how much they are meant to count in the objective, and when the two disagree the iterates converge to a prevalence-weighted surrogate rather than to the target risk. The gap is a bias, so no amount of training removes it. A minibatch is a partial-participation event over groups, which makes group-fair training an instance of federated aggregation under availability skew, and we carry the masked optimal transport construction of FedAVOT across. Aligning the batch-composition distribution q with the target importance distribution p produces per-batch group weights whose induced stochastic subgradient is exactly unbiased for the target risk, giving the standard $O(1/\sqrt{T})$ rate in the nonsmooth convex regime with a constant that does not depend on how many groups a batch happens to cover. Exact alignment is feasible precisely under a Hall-type condition, which reduces to the per-group requirement that importance not exceed observation rate; past it Sinkhorn returns an information projection and a residual we compute in closed form rather than bound. That closed form turns the excess risk into two terms, an alignment gain fixed by the instance and an optimization residual paid by the estimator, and a severity sweep over eleven availability skews on Adult and IMDB-Wiki shows the two crossing: transport wins while the gain exceeds the residual and loses past it, at a threshold the decomposition locates in advance. The residual is a stepsize artifact, not a property of the method, and a decaying schedule removes it on every instance where it had reversed the sign. A mean-CVaR layer composed on top reaches the worst group at rate $O(\kappa(\alpha, \gamma)/\sqrt{T})$.

Index Terms— Group fairness, optimal transport, stochastic gradient descent, distribution shift, conditional value at risk.

1. INTRODUCTION

A classifier is rarely asked to minimize one risk. It is asked to minimize one while behaving acceptably on subpopulations the training distribution under-represents: a diagnostic model across patient groups, a recognition system across demographic strata [1, 2, 3]. The standard response is to change the objective, replacing the empirical risk by $F_p(\theta) = \sum_i p_i f_i(\theta)$ with p the weighting one actually wants, uniform, inverse to prevalence, or externally dictated [4, 5].

This paper is about a gap between writing that objective down and optimizing it. Minibatch SGD does not see p : it sees a batch, whose group composition is governed by prevalence and by the sampler, so averaging the per-example gradients descends a different function. The estimator is not merely noisy, it is biased, by a fixed vector that survives $T \rightarrow \infty$ and any stepsize schedule. Rebalancing

the sampler is the usual and only partial fix: a group whose prevalence is below the resolution of the batch size cannot be sampled into every batch however the sampler is tilted, and upscaling the few examples that do appear reintroduces an amplitude distortion that destabilizes the iterates [6].

Groups as clients. The same mismatch is the central object of a different literature. In federated learning the *availability distribution* q over which clients participate in a round rarely matches the *importance distribution* p that defines the objective, and FedAvg consequently optimizes a surrogate whose weights are induced by the availability process rather than chosen [7, 8, 9]. FedAVOT [6] removes that mismatch by posing aggregation as a masked optimal transport (MOT) problem: find a coupling whose column marginal is the availability law and whose row marginal is the target p , supported only on client-round pairs that actually occurred. The observation driving this paper is that a minibatch is a participation event: the groups represented in B^t are the active set S^t , its law is q , and the weighting one wants is p . Group-fair SGD is federated aggregation with the federation collapsed onto one machine. We call the method GAVOT and keep the FedAVOT label in figures, the aggregation rule being unchanged.

Contributions.

1. *The reduction* (Sec. 2). Group-blind minibatch SGD reduces, under uniform-within-batch averaging, to agnostic FedAvg with one local step, so the surrogate identity of [7] names the function SGD actually minimizes, and transport weights make the batch subgradient exactly unbiased for ∇F_p (Lemma 5).
2. *A coverage law and a severity measure* (Cor. 4). Feasibility is Hall-type and reduces on singletons to $p_i \leq r_i$, importance below observation rate, so the *infeasible mass* $\nu \triangleq \sum_{i: p_i > r_i} p_i$ is the natural severity axis, computable before training.
3. *A two-term decomposition that locates the crossover* (Thm. 8, Sec. 5). Past the Hall condition the excess splits into an *alignment gain* Δ_{floor} , the distance between the two optima each rule actually converges to, and an optimization residual. The gain is exact and instance-fixed; the residual is paid by the estimator. We sweep eleven skews and watch the two cross.
4. *Risk-averse layer* (Sec. 4). Mean-CVaR over groups composes with the transport weights without disturbing unbiasedness, at rate $O(\kappa(\alpha, \gamma)/\sqrt{T})$, $\kappa(\alpha, \gamma) = \gamma + (1 - \gamma)/\alpha$.

Related work. Reduction methods for group fairness return randomized mixtures or carry an approximation constant tied to a fixed class [11, 12, 13]. Distributionally robust and worst-group objectives [3, 2, 28] change the functional rather than the sampler and are complementary: our CVaR layer is one such functional, obtained on top of transport rather than instead of it. Importance sampling for

SGD [16, 17] corrects variance, not the identity of the minimized objective.

2. PROBLEM FORMULATION

Groups and objectives. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a feature space and $\mathcal{Y}$ a label space, and let the population be partitioned into N groups indexed by $i \in [N]$ (demographic categories, devices, environments) with prevalences $\pi \in \Delta^{N-1}$, $\pi_i = \mathbb{P}[\text{group } i]$. Group i carries a distribution $\mathcal{D}_i$ on $\mathcal{X} \times \mathcal{Y}$ and the risk

$$f_i(\theta) \triangleq \mathbb{E}_{(X,Y) \sim \mathcal{D}_i} [\ell(m(X; \theta), Y)], \quad \theta \in \mathcal{C} \subseteq \mathbb{R}^{d'},$$

where $m(\cdot; \theta)$ is the predictor and ℓ the loss. Empirical risk minimization weights the groups by prevalence, $F_\pi = \sum_i \pi_i f_i$, and a majority group therefore dominates the trained model. The remedy is to fix an *importance distribution* $p \in \Delta^{N-1}$ and minimize

$$F_p(\theta) \triangleq \sum_{i=1}^N p_i f_i(\theta), \quad \theta^* \in \arg \min_{\theta \in \mathcal{C}} F_p(\theta). \quad (1)$$

Uniform p is the category-uniform objective, $p_i \propto 1/\pi_i$ the inverse-prevalence one, and p may equally encode a fairness constraint or be imposed externally [4, 5]. Throughout,

Assumption 1. $\mathcal{C}$ is convex and compact with diameter D ; each f_i is convex on $\mathcal{C}$ and $0 \leq f_i \leq M$; and every stochastic subgradient $g_i(\theta)$ of f_i returned by the sampler satisfies $\mathbb{E} \|g_i(\theta)\|^2 \leq G^2$ on $\mathcal{C}$.

These are the assumptions of [7, 6] restated for groups; they place no restriction on p or on π .

Prevalence versus importance. Minibatch SGD does not see p . At step t the sampler draws a batch B^t of b examples; write

$$A^t \triangleq \{i \in [N] : B^t \text{ contains an example from group } i\}$$

for the *active set* of groups, let $\{A_j\}_{j \in [M]}$ enumerate its support, and let $q_j \triangleq \mathbb{P}[A^t = A_j]$ be the *composition law*. The observation rate of group i is $r_i \triangleq \mathbb{P}[i \in A^t] = \sum_{j: i \in A_j} q_j$; under i.i.d. sampling $r_i = 1 - (1 - \pi_i)^b$, so a group whose prevalence is below the resolution of the batch is absent from most batches whatever p says. This is federated learning under partial participation: A^t is the set of participating clients, q the availability distribution, p the importance distribution [6, 7]. We treat q as known; otherwise it is estimated offline by simulating the sampler, as in our experiments.

What group-blind SGD minimizes. The group-blind update averages the per-example subgradients of B^t , which assigns group $i \in A^t$ the weight $\hat{\pi}_i^t$, its share of the batch, and assigns absent groups zero. Averaging first over the batch given A^t and then over $A^t \sim q$ is the hierarchical experiment of [7, Eq. 4], and it names the function SGD is in fact minimizing.

Proposition 2 (Surrogate objective). *Under Assumption 1, group-blind minibatch SGD is a stochastic subgradient method on $F_{\tilde{p}}$ with*

$$\tilde{p}_i = \sum_{j: i \in A_j} q_j \mathbb{E}[\hat{\pi}_i^t | A^t = A_j], \quad (2)$$

a probability vector. For i.i.d. sampling $\tilde{p} = \pi$; for uniform weighting within the batch $\tilde{p}_i = \sum_{j: i \in A_j} q_j / |A_j|$, the surrogate of agnostic FedAvg [6, Eq. 2]. Consequently, for every $\theta \in \mathcal{C}$,

$$|F_p(\theta) - F_{\tilde{p}}(\theta)| \leq M \|p - \tilde{p}\|_1, \quad (3)$$

and any method that minimizes $F_{\tilde{p}}$ exactly incurs the excess $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq 2M \|p - \tilde{p}\|_1$ on (1), which does not vanish with T .

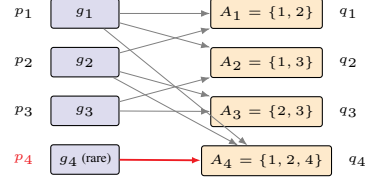


Fig. 1. The mask E . Batch compositions carry the mass q they occur with; groups must receive the mass p the objective assigns them. Group 4 is reachable only through A_4 , so exact alignment requires $p_4 \leq q_4$: the Hall condition of Thm. 3 read on the singleton $I = \{4\}$. Shrinking the batch deletes A_4 from the support and makes p_4 unreachable.

Proof. Conditionally on $A^t = A_j$ the within-group averages are unbiased for ∂f_i , so $\mathbb{E}[\text{batch subgradient} | A^t = A_j] = \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t | A_j] \partial f_i(\theta)$; averaging over q gives $\sum_i \tilde{p}_i \partial f_i = \partial F_{\tilde{p}}$, and $\sum_i \tilde{p}_i = \sum_j q_j \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t | A_j] = 1$. Both bounds follow from $0 \leq f_i \leq M$ and the two-point inequality $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq [F_p - F_{\tilde{p}}](\theta_{\tilde{p}}^*) + [F_{\tilde{p}} - F_p](\theta^*)$. $\square$

Proposition 2 says that the choice of p in (1) is decorative unless the aggregation step is changed: the iterates converge, at the usual rate, to the minimizer of a function whose weights are induced by the sampler rather than chosen, and the bias $p - \tilde{p}$ survives every step-size schedule. Rebalancing the sampler is only a partial fix: $\tilde{p}_i \leq r_i$ always, so a group with $p_i > r_i$ cannot be sampled into enough batches however the sampler is tilted, while upscaling the examples that do appear reintroduces an amplitude distortion that destabilizes the iterates [6]. The question is the one posed for federated aggregation in [6], read on batches:

Can the per-batch group weights be chosen so that SGD driven by the composition law q converges for the objective defined by p ?

The next section answers it, and characterizes exactly when the answer is yes.

3. PROPOSED APPROACH: ALIGNMENT BY MASKED OPTIMAL TRANSPORT

Weights as a coupling. Let $Y[i, j] \geq 0$ be the weight assigned to group i when the active set is A_j , with $Y[i, j] = 0$ for $i \notin A_j$, and let $T[i, j] \triangleq q_j Y[i, j]$ be the joint mass carried from composition j to group i . Requiring that the weights be a probability vector on each realized composition, and that each group receive in expectation the mass p assigns it, is the *masked optimal transport* (MOT) feasibility problem of [6],

$$T \mathbf{1}_M = p, \quad \mathbf{1}_N^\top T = q, \quad T \geq 0, \quad \text{supp}(T) \subseteq E, \quad (4)$$

with mask $E = \{(i, j) : i \in A_j\}$ encoding that a group absent from the batch cannot be weighted. Unlike classical transport [25, 20] there is no cost to minimize: any coupling respecting (4) serves, and (4) is the Kantorovich problem with the indicator cost $C[i, j] = \infty$ off E . Fig. 1 is the picture.

When alignment is possible. Existence of a coupling is a supply-demand question on the bipartite graph E , and is settled by flow duality.

Algorithm 1 GAVOT: group-fair SGD by masked transport

Require: target p , composition law $(q, \{A_j\}, E)$, step η , horizon T , CVar parameters (α, γ) with $\gamma = 1$ for the risk-neutral method

```

1:  $(T, Y) \leftarrow$  Sinkhorn scaling for  $(p, q, E)$   $\triangleright$  offline, once
2:  $\theta^0 \in \mathcal{C}$ ,  $t^0 \in \mathbb{R}$ 
3: for  $t = 1, \dots, T$  do
4:   draw batch  $B^t$ ; set  $A^t$ , let  $j(t)$  be s.t.  $A_{j(t)} = A^t$ 
5:    $g_i^t \leftarrow$  stochastic subgradient of  $f_i$  from  $B^t \cap$  group  $i$ 
6:    $w_i^t \leftarrow Y[i, j(t)](\gamma + \frac{1-\gamma}{\alpha} \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$ 
7:    $\theta^t \leftarrow \Pi_{\mathcal{C}}(\theta^{t-1} - \eta \sum_{i \in A^t} w_i^t g_i^t)$ 
8:    $t^t \leftarrow t^{t-1} - \eta(1 - \gamma)(1 - \frac{1}{\alpha} \sum_{i \in A^t} Y[i, j(t)] \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$ 
9: end for
Ensure:  $\bar{\theta}_T = \frac{1}{T} \sum_{t=1}^T \theta^t$ 

```

Theorem 3 (Feasibility, [6, Thm. 1]). *For $I \subseteq [N]$ let $\mathcal{N}(I) = \{j : \exists i \in I, (i, j) \in E\}$ and $\mathcal{S}(I) = \{j : A_j \subseteq I\}$. Problem (4) is feasible if and only if*

$$\sum_{j \in \mathcal{S}(I)} q_j \leq \sum_{i \in I} p_i \leq \sum_{j \in \mathcal{N}(I)} q_j \quad \forall I \subseteq [N]. \quad (5)$$

Proofs are in the appendix. Read on singletons, (5) becomes a statement one can act on before training starts.

Corollary 4 (Coverage law and severity). *Feasibility of (4) requires $\mathbb{P}[A^t = \{i\}] \leq p_i \leq r_i$ for every i , with $r_i = \mathbb{P}[i \in A^t]$ the observation rate of Sec. 2. Writing*

$$\nu \triangleq \sum_{i: p_i > r_i} p_i \quad (6)$$

for the infeasible mass, $\nu = 0$ iff the upper constraints hold. Under i.i.d. sampling of b examples the upper constraint reads $b \geq \log(1 - p_i) / \log(1 - \pi_i)$; under K -of- N uniform group sampling it reads $p_i \leq K/N$.

Severity. Importance may not exceed observation rate: a rare group is reachable only if the batch is large enough to contain it often enough. The infeasible mass ν is the axis we sweep: at $\nu = 0$ exact alignment is available, and as ν grows more of p sits outside the set of row marginals the mask can reach. A *feasible* and an *infeasible* regime are two points on this continuum, and Sec. 5 sweeps it instead.

Computing the plan. Given feasibility, a coupling is obtained by iterative proportional fitting (Sinkhorn scaling) on the masked support [18, 19, 20]: starting from any $T^{(0)} > 0$ on E with $\mathbf{1}^\top T^{(0)} = q$, alternate

$$T \leftarrow \text{diag}(p \oslash T \mathbf{1}) T, \quad T \leftarrow T \text{diag}(q \oslash T^\top \mathbf{1}) \quad (7)$$

until $\|T \mathbf{1} - p\|_1 \leq \varepsilon$. Each sweep costs $O(|E|)$ with $|E| \leq NM$ and M the number of *realized* compositions rather than 2^N ; the plan is computed once, offline, and the training loop adds one lookup and one multiply per group per step. Under (5) the iteration converges to a solution of (4) [18, 22], being block-coordinate ascent on the dual of the entropic problem $\min\{\sum_E T \log T : T \mathbf{1} = p, \mathbf{1}^\top T = q\}$. Since $\mathbf{1}_N^\top T = q$ and $Y = T \text{diag}(q)^{-1}$, every column $Y[\cdot, j]$ is a probability vector on A_j , the property each bound below uses. Algorithm 1 is the resulting method; $\gamma = 1$ is the risk-neutral rule analyzed here and $\gamma < 1$ the risk-averse layer of Sec. 4.

Analysis. Everything rests on one identity, [6, Lem. 1] moved from model averaging to gradient averaging. Let $g_i(\theta)$ be the stochastic subgradient of f_i computed from $B^t \cap$ group i and $\hat{g}(\theta) = \sum_{i \in A^t} Y[i, j(t)] g_i(\theta)$ the transport-weighted batch subgradient of Algorithm 1. Transport removes bias; it is not a variance reduction device.

Lemma 5 (Unbiasedness). *Let T satisfy (4). Then $\mathbb{E}[\hat{g}(\theta) \mid \theta] \in \partial F_p(\theta)$ for every $\theta \in \mathcal{C}$.*

Proof. Conditionally on $A^t = A_j$, $\mathbb{E}[g_i(\theta)] \in \partial f_i(\theta)$ for $i \in A_j$. Averaging over $A^t \sim q$, $\mathbb{E}[\hat{g}] = \sum_j q_j \sum_{i \in A_j} Y[i, j] \partial f_i(\theta) = \sum_i (\sum_j T[i, j]) \partial f_i(\theta) = \sum_i p_i \partial f_i(\theta)$, the row-marginal constraint being used in the last step. $\square$

Lemma 6 (Second moment, free of coverage). $\mathbb{E}\|\hat{g}(\theta)\|^2 \leq G^2$ for every $\theta \in \mathcal{C}$, with no dependence on $|A^t|$ or on N .

Proof. $Y[\cdot, j]$ is a probability vector on A_j , so $\|\sum_i Y[i, j] g_i\|^2 \leq \sum_i Y[i, j] \|g_i\|^2$ by convexity of $\|\cdot\|^2$; take expectations and use $\sum_j T[i, j] = p_i$: $\mathbb{E}\|\hat{g}\|^2 \leq \sum_i p_i \mathbb{E}\|g_i\|^2 \leq G^2$. $\square$

Theorem 7 (Rate). *Under Assumption 1, if (4) is feasible then Algorithm 1 with $\gamma = 1$ and $\eta = D/(G\sqrt{T})$ satisfies*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) \leq \frac{DG}{\sqrt{T}}. \quad (8)$$

The constant in (8) is that of full-participation SGD and is independent of how many groups appear in a batch: a composition covering two groups yields the same bound as one covering all N , though with larger realized variance. The proof uses only the two lemmas and never refers to $|A^t|$, which is why the rate of [6] transfers verbatim from aggregation of models to aggregation of gradients.

Outside the feasible region. When (5) fails, (4) has no solution and the iteration (7) does not diverge: its iterates accumulate at the coupling whose row marginal $\hat{p} \triangleq T \mathbf{1}$ is the KL-closest to p over the reachable polytope $\mathcal{R}(q, E) = \{T \mathbf{1} : T \geq 0, \mathbf{1}^\top T = q, \text{supp}(T) \subseteq E\}$ [22, 21, 20]. The column marginal is met exactly, so Lemmas 5 and 6 hold with p replaced by $\hat{p}$: Algorithm 1 is then unbiased SGD on the surrogate $F_{\hat{p}}$, the closest objective the sampler can reach, and the loss on (1) is a bias of exactly the shape of Proposition 2. On singletons $\hat{p}_i \leq r_i$, so the groups with $p_i > r_i$ are truncated at their observation rate and their missing mass, ν up to renormalization, is redistributed.

Theorem 8 (Infeasible regime). *Let $\hat{p}$ be the row marginal of the Sinkhorn limit, $\theta^\dagger \in \arg \min_{\mathcal{C}} F_{\hat{p}}$, and write $\Phi(w) \triangleq F_p(\arg \min_{\mathcal{C}} F_w)$ for the p -risk of the minimizer of the w -weighted risk. Then*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) = \underbrace{\Phi(\hat{p}) - \Phi(p)}_{\text{alignment floor}} + \underbrace{\varepsilon_T(\hat{p})}_{\text{optimization residual}}, \quad (9)$$

where $\varepsilon_T(\hat{p}) \triangleq \mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^\dagger)$ satisfies $\mathbb{E}F_{\hat{p}}(\bar{\theta}_T) - F_{\hat{p}}(\theta^\dagger) \leq DG/\sqrt{T}$ and $\varepsilon_T(\hat{p}) \rightarrow 0$ whenever $\theta_T \rightarrow \theta^\dagger$. *Group-blind averaging obeys (9) with $\hat{p}$ in place of $\hat{p}$. Writing $\Delta_{\text{floor}} \triangleq \Phi(\hat{p}) - \Phi(\hat{p})$ for the alignment gain, transport improves on group-blind averaging exactly when $\Delta_{\text{floor}} > \varepsilon_T(\hat{p}) - \varepsilon_T(\hat{p})$, and every floor is bounded by $\Phi(w) - \Phi(p) \leq 2M\|p - w\|_1$.*

Theorem 8 replaces the usual bias *bound* [6] by an exact *decomposition*, both terms of which are available before training: $\hat{p}$ and $\hat{p}$ from (p, q, E) , and Φ exactly on instances with a closed-form or convex frontier. The gain is fixed by the instance and decides the sign of the comparison; the residual is paid by the estimator and, as Sec. 5 shows, is a stepsize artifact. The ℓ_1 bound is the weaker object and we report both, because it turns out too loose to discriminate.

4. A RISK-AVERSE LAYER OVER GROUPS

Exact alignment delivers the p -weighted risk, an average, which can hide a group. Where the requirement is on the worst group instead, we compose a mean-CVaR functional over groups on top of the transport weights [7, 26]:

$$F_p^{\alpha, \gamma}(\theta) \triangleq \gamma F_p(\theta) + (1 - \gamma) \text{CVaR}_\alpha^p(f(\theta)), \quad \alpha, \gamma \in (0, 1], \quad (10)$$

where CVaR_α^p is the conditional value at risk of the group-loss vector under p . The Rockafellar-Uryasev representation turns (10) into a joint minimization over (θ, t) ,

$$\Phi(\theta, t) = \gamma \sum_i p_i f_i(\theta) + (1 - \gamma) \left(t + \frac{1}{\alpha} \sum_i p_i [f_i(\theta) - t]_+ \right), \quad (11)$$

jointly convex, with subgradients in both arguments again p -weighted sums of per-group quantities, so Lemma 5 applies unchanged. This is why the layers compose rather than interfere: transport fixes *which* distribution the expectation is under, CVaR changes *what* is averaged. Lines 6 and 8 of Algorithm 1 implement (11).

Theorem 9 (Rate under mean-CVaR). *Under the hypotheses of Theorem 7, Algorithm 1 with $\eta = \Theta(1/\sqrt{T})$ satisfies $\mathbb{E}[F_p^{\alpha, \gamma}(\theta_T)] - \min F_p^{\alpha, \gamma} = O(\kappa(\alpha, \gamma)/\sqrt{T})$ with $\kappa(\alpha, \gamma) = \gamma + (1 - \gamma)/\alpha$. In the infeasible regime the bound of Theorem 8 acquires the same factor on its statistical term, the bias term being unchanged.*

Both $\gamma = 1$ and $\alpha = 1$ recover Theorem 7 exactly, so the grid contains the risk-neutral method along two edges, visible as the constant row and column of every heatmap below. Sending $\alpha \downarrow 0$ drives (10) to the worst group at the cost $\kappa(\alpha, \gamma) \rightarrow \infty$: the parameter prices robustness in the convergence constant, not in the limit.

Remark 10 (Relation to constrained learning). $\max_i f_i$ is the exact-penalty limit of $\min f_0$ s.t. $f_i \leq c_i$, a fixed price on the largest violation matching penalized and constrained values once it exceeds the least ℓ_1 norm on the dual-optimal set [15]. The $\alpha \downarrow 0$ edge of (10) is thus the scalarization a constrained specification selects, with κ its price in the rate.

5. EXPERIMENTS

Instances. *Adult* income classification [33]: logistic regression ($d=109$ with intercept, cross-entropy, $\eta=0.1$) on 100 race-homogeneous groups of 30 samples, the number of groups per race proportional to its prevalence (White/Black/Asian-Pac./Amer-Ind./Other = 85/10/3/1/1); the importance target is the fairness target itself, 1/5 per race divided equally among its groups, so a group from the smallest race carries $p_i = 0.2$. *IMDb-Wiki* age regression [35]: linear regression of age on 128-d ResNet face embeddings ($\eta=10^{-2}$) over the first 100 identities with 30 images, with the cubic importance target $p_i \propto (101 - i)^3$. Observation rates are swept through a one-parameter family: on Adult $r \propto p^\beta$, $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, from uniform observation (prevalence regime, $\nu=.60$: the five groups of the three smallest races hold 60% of the target against $r_i \approx K/m$) to importance-aligned ($\nu=0$); on IMDb-Wiki $r_i \propto i^\beta$, $\beta \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$, from uniform observation ($\nu=.31$, nine groups) to the mirror image of p ($\nu=.88$, 41 groups), plus the aligned $r_i \propto 101 - i$ ($\nu=0$). Eleven instances, ν from 0 to .88.

Protocol. Each step observes $K=3$ groups drawn without replacement with probabilities $\propto r$, which induces q over the $\binom{100}{3}$

Table 1. Headline cells, tail-500 over 5 seeds, decaying stepsize. *full* sees every group every step; *upsamp.*, self-normalized upsampling (weights $\propto p_i/r_i$ within the batch); LDS, label-distribution smoothing [34], regression only. The fixed multiplier m/K diverges whenever observation rates are non-uniform.

Instance	GAVOT	unif. avg	upsamp.	LDS	m/K	full
Adult, $\nu=.60$	0.2269	0.2479	0.2288	—	0.670	0.2073
Adult, $\nu=0$	0.2075	0.2077	0.2075	—	div.	0.2073
IMDb, $\nu=.31$	86.06	91.63	86.74	94.28	8504	82.95
IMDb, $\nu=0$	84.28	86.37	84.40	89.43	div.	82.95

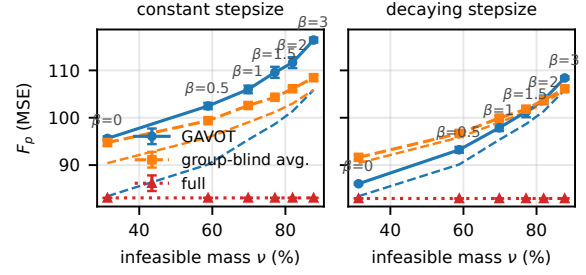


Fig. 2. IMDb-Wiki severity sweep. Solid, measured tail-500 (± 1 std); dashed, the exact floor $\Phi(\cdot)$ each rule converges to. Left, constant stepsize: the optimization residual swamps the alignment gain and transport loses everywhere, even where the floors are seven units apart. Right, decaying stepsize: the residual collapses and the measured curves track their floors, so transport wins until the floors close at high ν . The sign is set by which of the two terms of (9) dominates, not by the regime label.

compositions; q is estimated from 10^6 draws of the sampler and the plan from (7) on the membership mask, once per instance. Each observed group takes $H=5$ local steps on its own data before the weighted average, the standard practical deviation from the one-step analysis, applied to every rule alike since all rules consume the same local updates from the same draw A^t . Rules: GAVOT; *group-blind averaging*, the uniform average over the observed groups (Proposition 2 with $\tilde{p}_i = r_i/K$); the *fixed multiplier* m/K , the rescaling $(m/K)p_i$ that presumes uniform observation; and *full*, every group every step weighted by p , the reference $\Phi(p)$. Stepsize $\eta_t = \eta/(1 + t/1000)$ unless stated, fixed before the comparison and shared by every rule; across nine schedules (constant, hyperbolic with $T_0 \in \{250, \dots, 16000\}$, inverse square root, step, cosine) the winner never changes on any instance, only the margin does: a faster decay puts full coverage on its floor and widens the IMDb-Wiki gap (cosine: 84.12 against 90.69), a constant stepsize narrows it to a tie. 4000 steps, 5 seeds; we report F_p averaged over the last 500 steps, mean $\pm$ std over seeds. On both instances $\arg \min F_w$ is available to machine precision (closed form for least squares, Newton for the logistic loss), so the floors $\Phi(p)$, $\Phi(\tilde{p})$, $\Phi(\tilde{p})$ of Theorem 8 are exact rather than estimated, and in every instance with $\nu = 0$ the iteration (7) converges to a row error below $3 \cdot 10^{-8}$, certifying the remaining inequalities of (5).

The least represented group. Fig. 3 reads the same runs on the group the target exists for. On Adult, transport lowers the loss on race Other at every skew and under both stepsizes: .073 against .119 for group-blind averaging on the prevalence instance (full coverage .031), and .031 against .035 on the aligned one, where it matches

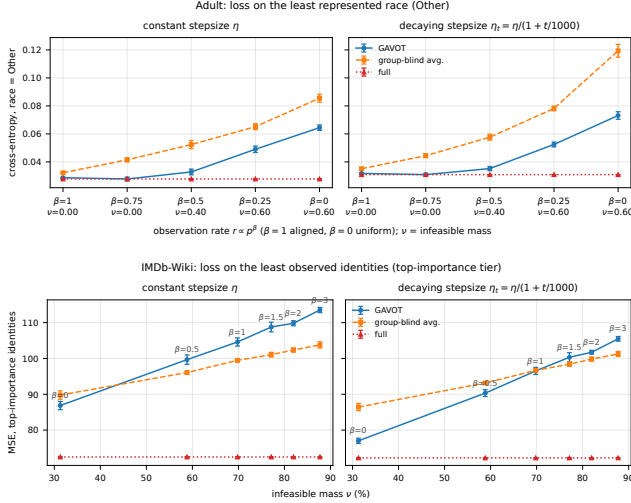


Fig. 3. Loss on the least represented group, the metric the fairness target is about. Top, Adult: cross-entropy on race Other, one group of 30 out of 100, to which the category-uniform target assigns $p_i = 0.2$; the axis is the observation-rate exponent, $\beta=1$ aligned, $\beta=0$ uniform (prevalence). Bottom, IMDb-Wiki: MSE on the 20 highest-importance identities, the least observed ones once $\beta > 0$, against ν . Left panels constant stepsize, right panels decaying. Same protocol as the tables ($K=3$, $H=5$, 4000 steps, tail-500 mean $\pm$ std over 5 seeds); lower is better. Curves are the group-wise entries of the same runs as Table 1 and Fig. 2.

full coverage. Amer-Indian, the other single-group race, behaves the same way. On IMDb-Wiki the top-importance identities show the crossover of Table 2 one step earlier: under decay transport wins at $\nu \leq .70$ (77.0 against 86.4 at $\nu=.31$, full 72.2; 73.6 against 79.9 when aligned) and loses at the three most infeasible skews, while at constant stepsize it wins only at $\nu=.31$. The single most starved group, the one with the largest p_i/r_i in each instance, sharpens the picture: on Adult it is race Other itself ($p_i/r_i=6.7$); on IMDb-Wiki it is the top identity, where under decay transport gives 253 against 331 at $\nu=.31$ and 373 against 380 at $\nu=.59$ (full 219), then loses from $\nu=.70$ on, the same crossover one step earlier still (a 30-image group, so ± 5 to 10 across seeds). The overall objective and the rare-group loss therefore tell the same story, with the rare group the more sensitive of the two.

Why transport stops helping. The failure is not a bug of the weights; it is the two terms of (9) changing sign. A starved group has $p_i > r_i$, so the plan can deliver at most r_i of its mass: past the coverage law of Cor. 4 the reachable target $\hat{p}$ truncates at the observation rate, and as ν grows $\hat{p}$ and the group-blind $\tilde{p}$ starve the same groups, so the two floors converge (Table 2, Δ_{floor} from 6.99 to 0.02). What transport still does is concentrate weight on the starved group whenever it does appear, p_i/r_i times its batch share, which is exactly the variance Lemma 6 does not see: the second moment is bounded, the variance about the mean is not, and it is largest on the rare group. Once the floors coincide that variance is all that separates the two rules, so transport loses by the residual, by 2 to 8 at constant stepsize and by 2.3 at $\nu=.88$ even under decay. Transport helps exactly while there is mass left to move.

E1: the headline cells. Table 1 reports the four instances a practitioner would recognize. Transport wins on three: Adult under

Table 2. IMDb-Wiki severity sweep, decaying stepsize. Δ_{floor} is the alignment gain of Theorem 8, *gain* the measured advantage of transport, t its Welch statistic (population std). The excess residual transport pays averages 1.9, and the sign flips exactly where Δ_{floor} crosses it. The five Adult skews all sit above their crossover and are omitted.

β	ν	Δ_{floor}	gain	t
0	.31	6.99	+5.57	20.7
0.5	.59	5.71	+3.46	11.3
1.0	.70	3.67	+2.05	5.4
1.5	.77	2.56	+0.58	1.1
2.0	.82	1.62	−0.03	−0.1
3.0	.88	0.02	−2.28	−7.9

prevalence-proportional availability, 0.2269 against 0.2479 ($t=45$), and both IMDb cells, 86.06 against 91.63 ($t=21$) and 84.28 against 86.37 ($t=13$). On Adult with importance-aligned availability the two coincide at the oracle value, which is what Proposition 2 predicts when $\tilde{p} = p$: there is nothing to correct and transport does not manufacture a difference. The fixed multiplier diverges on both aligned instances and is far off on the other two, so naive rescaling is no alternative to solving (4). Two imbalance baselines sit between the two. Self-normalized upsampling (weights $\propto p_i/r_i$, renormalized within the batch), the reweighting form of oversampling rare groups, recovers most of the gain and transport the rest: 0.2288 against 0.2269 on Adult prevalence ($t=5.8$) and 86.74 against 86.06 on IMDb skewed ($t=2.6$, at the margin of significance with 5 seeds); on the aligned instances the two coincide, as every convex rule must. Downsampling ($\min(p_i/r_i, 1)$, normalized) is never better than upsampling (0.2308, 86.90). Label-distribution smoothing [34], the standard imbalanced-regression correction, reweights by the rarity of the label value rather than by how rarely a group is observed, and trails even group-blind averaging on both IMDb-Wiki instances (94.28, 89.43). On the least represented group the margin of transport over upsampling widens: 77.0 against 80.0 at $\nu=.31$ and 73.6 against 75.6 when aligned on IMDb-Wiki, .073 against .075 on Adult prevalence. Upsampling is the one-shot heuristic for the marginal problem (4) solves exactly, without the feasibility certificate or the floor decomposition.

E2: the crossover, and where the sign comes from. Table 2 and Fig. 2 are the paper’s main evidence. On Adult, whose five skews are omitted from the table, the measured gain tracks Δ_{floor} within 4% at three of five (.0134 against .0129, .0050 against .0049, .0002 against .0003) and exceeds it at the two most infeasible (.0210 against .0110, .0181 against .0144), group-blind averaging sitting further above its floor. On IMDb-Wiki the gain is uniformly below the floor by an excess optimization residual that is close to constant in β , mean 1.9 in MSE units, and the measured sign flips precisely where Δ_{floor} crosses that value, between $\beta=1.5$ (2.56, gain +0.58) and $\beta=2.0$ (1.62, gain −0.03). This is Theorem 8 read at finite T : the sign of the comparison is a quantity one computes in advance.

E3: the ℓ_1 bound is too loose to be the criterion. The distances $\|p - \hat{p}\|_1$ and $\|p - \tilde{p}\|_1$ are also computed on every instance, and $\|p - \hat{p}\|_1 < \|p - \tilde{p}\|_1$ holds in all eleven, including the three where transport loses or ties. The ℓ_1 form of Theorem 8 therefore never predicts a loss and cannot serve as the criterion; the exact floors can.

E4: the residual is a stepsize artifact. The left panel of Fig. 2 runs the identical instances at constant stepsize. There transport loses on all six IMDb skews, by as much as 7.9, even at $\nu=.31$ where the

floors are 6.99 apart. The transport weights $Y[\cdot, j]$ concentrate mass on rarely observed groups, so although Lemma 6 bounds the second moment identically for both rules, the variance about the mean is larger, and at constant stepsize the iterates sit on a noise floor proportional to it. Under $\eta_t = \eta/(1 + t/1000)$ that floor decays, the measured curves fall onto their dashed floors, and every sign reversal attributable to the stepsize disappears. The residual is a property of the schedule, not of the estimator.

E5: the decomposition in closed form. The mirrored IMDb-Wiki instance ($\nu=.88$) is where Theorem 8 can be read off exactly. The Sinkhorn limit truncates the target at the observation-rate ceiling, $\hat{p}_i \approx \min(p_i, r_i)$ up to renormalization (maximum deviation .023), with $\|p - \hat{p}\|_1 = 1.58$ of a possible 2. The frontier is least squares, so $\Phi(p) = 82.9$, matching the measured full plateau 82.95, and $\Phi(\hat{p}) = 105.9$: the alignment floor is 23.0 and the realized bias equals its exact expression $|(p - \hat{p})^\top L(\theta^\dagger)| = 30.7$, where L collects the per-group risks, while the ℓ_1 form $2M\|p - \hat{p}\|_1$ is conservative since $M \geq \max_i L_i(\theta^\dagger) \approx 454$. Group-blind averaging has the same floor to two decimals, $\Phi(\bar{p}) = 105.9$: both surrogates starve the same groups, so Δ_{floor} vanishes and the entire measured gap between the two rules at this endpoint of Table 2 is residual, which is why it is the one instance where transport still loses by a significant margin under decay ($\beta=2$, where the floors are 1.6 apart, is a tie).

E6: risk aversion, and where it does not help. Sweeping (α, γ) on a 10×10 grid, the overall objective is minimized on the $\gamma = 1$ edge on every instance, which is $\kappa(1, \gamma) = 1$ made visible and confirms that risk aversion costs the average. The one place the CVaR layer earns its keep is the worst race on Adult under prevalence availability, where GAVOT+CVaR at $(0.2, 0.7)$ reaches 0.3624 against 0.3679 for GAVOT and 0.3994 for group-blind averaging. Per race, transport improves four of five (Black .179 vs .200, Asian-Pac .368 vs .399, Am-Ind .175 vs .189, Other .073 vs .119) and regresses only the majority group (.339 vs .332), which is what reweighting toward a category-uniform target is supposed to do.

Why risk aversion does not reach the rare group. On the least represented group the CVaR layer never helps: over the whole (α, γ) grid the best rare-group loss is at $\gamma=1$, i.e. with the layer switched off, on all four headline instances, and at $(\alpha, \gamma) = (0.1, 0.1)$ it nearly triples the loss on race Other (.207 against .073) and raises the top-identity loss on IMDb-Wiki from 77.0 to 103. The reason is where the tilt lands. CVaR upweights the groups whose realized loss exceeds the running threshold, but it can only tilt among the groups in the batch, and the rare group is absent from most batches: the extra weight goes to whichever observed group is doing badly that step, not to the group the target is about. Composed with transport, the hinge multiplies weights that are already p_i/r_i large, so the product amplifies the variance of (9) rather than the alignment, which is why the loss grows as α shrinks. Risk aversion changes the functional; it does not change who is observed, and observation is what the rare group lacks. Transport moves mass toward it whenever it appears; CVaR cannot, and the one gain it produces, on the worst race above, comes from tilting among the groups that are present.

6. CONCLUSION

Group-fair training fails in a specific and fixable place: the aggregation step, blind to the weighting the objective declares. Reading a minibatch as a partial-participation event makes the fix the masked transport problem that corrects availability skew in federated learning, with the same feasibility characterization, an $O(1/\sqrt{T})$ guarantee free of batch coverage, and a coverage law whose infeasible mass ν orders instances by severity. Past feasibility the excess splits

cleanly, into an alignment gain fixed by the instance and an optimization residual paid by the estimator, and a sweep over eleven skews shows the two crossing where the decomposition says they will. Transport is worth applying while the gain exceeds the residual, both are computable before training, and the residual is a stepsize artifact a decaying schedule removes. The nonconvex case remains open.

7. REFERENCES

- [1] J. Buolamwini and T. Gebru, “Gender shades: Intersectional accuracy disparities in commercial gender classification,” in *Proc. Conf. Fairness, Accountability and Transparency (FAT*)*, 2018, pp. 77–91.
- [2] T. Hashimoto, M. Srivastava, H. Namkoong, and P. Liang, “Fairness without demographics in repeated loss minimization,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 1929–1938.
- [3] S. Sagawa, P. W. Koh, T. Hashimoto, and P. Liang, “Distributionally robust neural networks for group shifts,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.
- [4] M. Mohri, G. Sivek, and A. T. Suresh, “Agnostic federated learning,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 4615–4625.
- [5] T. Li, M. Sanjabi, A. Beirami, and V. Smith, “Fair resource allocation in federated learning,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.
- [6] H. Rahimi and D. Kalogerias, “FedAVOT: Exact distribution alignment in federated learning via masked optimal transport,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [7] H. Rahimi and D. Kalogerias, “Convergence of agnostic federated averaging,” in *Proc. IEEE Int. Workshop Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, 2025.
- [8] J. Wang, Q. Liu, H. Liang, G. Joshi, and H. V. Poor, “Tackling the objective inconsistency problem in heterogeneous federated optimization,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 7611–7623.
- [9] Y. J. Cho, J. Wang, and G. Joshi, “Towards understanding biased client selection in federated learning,” in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2022, pp. 10351–10375.
- [10] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.
- [11] A. Agarwal, A. Beygelzimer, M. Dudík, J. Langford, and H. Wallach, “A reductions approach to fair classification,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 60–69.
- [12] A. Cotter, H. Jiang, and K. Sridharan, “Two-player games for efficient non-convex constrained optimization,” in *Proc. Int. Conf. Algorithmic Learning Theory (ALT)*, 2019, pp. 300–332.
- [13] L. F. O. Chamon and A. Ribeiro, “Probably approximately correct constrained learning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 16722–16735.
- [14] L. F. O. Chamon, S. Paternain, M. Calvo-Fullana, and A. Ribeiro, “Constrained learning with non-convex losses,” *IEEE Trans. Inf. Theory*, vol. 69, no. 3, pp. 1739–1760, 2023.
- [15] H. Rahimi, S. Pougkakiotis, and D. Kalogerias, “An exact penalty method for constrained statistical learning over universal RKHSs,” 2026, submitted.
- [16] M. Rizk, S. Vlaski, and A. H. Sayed, “Optimal importance sampling for federated learning,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 3110–3114.
- [17] A. Katharopoulos and F. Fleuret, “Not all samples are created equal: Deep learning with importance sampling,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2018, pp. 2525–2534.
- [18] R. Sinkhorn, “Diagonal equivalence to matrices with prescribed row and column sums,” *Amer. Math. Monthly*, vol. 74, no. 4, pp. 402–405, 1967.
- [19] M. Cuturi, “Sinkhorn distances: Lightspeed computation of optimal transport,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 26, 2013, pp. 2292–2300.
- [20] G. Peyré and M. Cuturi, *Computational Optimal Transport*. Now Publishers, 2019.
- [21] C. Gietl and F. P. Reffel, “Accumulation points of the iterative proportional fitting procedure,” *Statistics*, vol. 47, no. 6, pp. 1321–1335, 2013.
- [22] I. Csiszár, “ I -divergence geometry of probability distributions and minimization problems,” *Ann. Probab.*, vol. 3, no. 1, pp. 146–158, 1975.
- [23] P. Hall, “On representatives of subsets,” *J. London Math. Soc.*, vol. 10, no. 1, pp. 26–30, 1935.
- [24] L. R. Ford and D. R. Fulkerson, “Maximal flow through a network,” *Canad. J. Math.*, vol. 8, pp. 399–404, 1956.
- [25] C. Villani, *Optimal Transport: Old and New*. Springer, 2009.
- [26] R. T. Rockafellar and S. Uryasev, “Optimization of conditional value-at-risk,” *J. Risk*, vol. 2, no. 3, pp. 21–41, 2000.
- [27] S. Curi, K. Y. Levy, S. Jegelka, and A. Krause, “Adaptive sampling for stochastic risk-averse learning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1036–1047.
- [28] D. Levy, Y. Carmon, J. C. Duchi, and A. Sidford, “Large-scale methods for distributionally robust optimization,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 8847–8860.
- [29] H. Namkoong and J. C. Duchi, “Variance-based regularization with convex objectives,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 2971–2980.
- [30] A. Nemirovski, A. Juditsky, G. Lan, and A. Shapiro, “Robust stochastic approximation approach to stochastic programming,” *SIAM J. Optim.*, vol. 19, no. 4, pp. 1574–1609, 2009.
- [31] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” in *Proc. Machine Learning and Systems (MLSys)*, 2020, pp. 429–450.
- [32] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, “SCAFFOLD: Stochastic controlled averaging for federated learning,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2020, pp. 5132–5143.
- [33] B. Becker and R. Kohavi, “Adult,” UCI Machine Learning Repository, 1996.
- [34] Y. Yang, K. Zha, Y.-C. Chen, H. Wang, and D. Katabi, “Delving into deep imbalanced regression,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2021.
- [35] R. Rothe, R. Timofte, and L. Van Gool, “Deep expectation of real and apparent age from a single image without facial landmarks,” *Int. J. Comput. Vis.*, vol. 126, no. 2–4, pp. 144–157, 2018.
- [36] P. Theodoropoulos, K. E. Nikolakakis, and D. Kalogerias, “Federated learning under restricted user availability,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2024.